Along the solutions of (1.3), the energy is conserved, since its derivative vanishes:

$$\frac{\mathrm{d}}{\mathrm{d}t}E = \sum_{j=1}^{\mathrm{N}} m_j \dot{\boldsymbol{q}}_j \cdot \ddot{\boldsymbol{q}}_j + \sum_{j=1}^{\mathrm{N}} \frac{\partial U}{\partial \boldsymbol{q}_j} \cdot \dot{\boldsymbol{q}}_j = \sum_{j=1}^{\mathrm{N}} \left(m_j \ddot{\boldsymbol{q}}_j + \frac{\partial U}{\partial \boldsymbol{q}_j} \right) \cdot \dot{\boldsymbol{q}}_j = 0.$$

As will become apparent in later chapters, it is sometimes necessary to introduce additional terms into the model to account for environmental conditions; these may destroy or alter the conservation of energy.

When we refer to the *value* of energy in a conservative system we typically write E instead of E (which represents the energy function).[3] Thus the positions and velocities of system evolving under Newtonian dynamics satisfy the equation

$$E(\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_{\mathrm{N}}, \dot{\boldsymbol{q}}_1, \dot{\boldsymbol{q}}_2, \ldots, \dot{\boldsymbol{q}}_{\mathrm{N}}) = \mathrm{E}.$$

In addition to the energy, there may be other constants of motion which arise from the structure of the equations of motion. For example, if the full set of particles of an N-body system is included in the model, then the instantaneous force acting on particle i due to the presence of particle j will be the exact negative of the force acting on particle j due to particle i; this is a direct consequence of Newton's third law (every action has an equal and opposite reaction). This means that the sum of all the forces will vanish. Since

$$\frac{\mathrm{d}\boldsymbol{p}_i}{\mathrm{d}t} = \boldsymbol{F}_i,$$

we have

$$\sum_{i=1}^{\mathrm{N}} \frac{\mathrm{d}\boldsymbol{p}_i}{\mathrm{d}t} = \sum_{i=1}^{\mathrm{N}} \boldsymbol{F}_i = 0$$

and thus the three components of the *total momentum vector* $\boldsymbol{p}_{\mathrm{tot}} := \sum_{i=1}^{\mathrm{N}} \boldsymbol{p}_i$ will be conserved quantities. Additional first integrals sometimes arise in specific situations, depending on the form of the potential energy function.

For a few very special N-body systems, analytical solution is possible.

Example 1.3 (Harmonic Oscillator) Consider the system

$$\dot{x} = v, \qquad \dot{v} = -\Omega^2 x.$$

[3] We use roman (non-italic) letters for parameters that describe the thermodynamic state, e.g. number of degrees of freedom, the volume, the energy, the temperature or the pressure.

This system describes the behavior of a particle with unit mass in one dimension, with energy function $E(x, \dot{x}) = \dot{x}^2/2 + \Omega^2 x^2/2$, where its motion is governed by a linear 2nd order equation $\ddot{x} + \Omega^2 x = 0$. The solution, for given $x(0) = \xi$, $\dot{x}(0) = v(0) = \eta$, is

$$x(t) = \xi \cos(\Omega t) + \frac{\eta}{\Omega} \sin(\Omega t).$$

This system, despite its simplicity, is an important test case for many schemes in molecular dynamics and will appear in many forms throughout this book (we generally refer to any system where the energy has quadratic position dependence as harmonic and any one degree of freedom harmonic system as a harmonic oscillator).

Example 1.4 (Single Degree of Freedom) Consider a simple system with a single degree of freedom (with unit mass $m = 1$ for simplicity) and energy function $E(x, \dot{x}) = \dot{x}^2/2 + U(x)$ (which includes the harmonic oscillator as a special case). The equations of motion are

$$\dot{x} = v, \qquad \dot{v} = -\frac{\partial U}{\partial x}.$$

Solutions are smooth functions $(x(t), v(t))$ that evolve on curves that are defined by $E(x, v) \equiv \mathrm{E}$, where E is the energy calculated at the initial point $(x(0), v(0)) = (\xi, \eta)$. Near this point, provided $\eta \neq 0$, let us solve the equation $E(x, v) = E(\xi, \eta)$ for v as a unique smooth function of x, ξ, η using the implicit function theorem: $v = V(x, \xi, \eta)$. Inserting this into the differential equations we then find

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v = V(x, \xi, \eta).$$

This is a separable differential equation; it is therefore easily integrated from any provided initial value, $x(0) = \xi$, as a function of t, resulting in a relation $x = X(t, \xi, \eta)$. Then $v = V(X(t, \xi, \eta), \xi, \eta)$ and we see that it is possible to derive the formula for the solution (x, v) as a function of t and the initial conditions. This representation provides a complete characterization of the *local* behavior of solutions, i.e., within a small interval of our starting point. To describe the *global* solutions, we would need to piece together several such local representations. For example if we used this approach to solve the harmonic oscillator, the equation for x would be $\dot{x} = \pm\sqrt{2\mathrm{E} - \Omega^2 x^2}$, and the sign that appears would define different parts of the solution which may be concatenated together to describe a global solution. "Degenerate" points, where $\partial E/\partial v = 0$, lead to difficulties that must be resolved on a case by case basis.

In more complicated systems, it is usually impossible to write down a formula for the solution in an explicit form; so approximation (typically using numerical methods) becomes essential to being able to predict dynamics.